bacterial enzymes that recognize a specific base sequence in a DNA molecule (from any source) and make two cuts one in each strand generating 3′ – OH and 5′ – P termini.

There are several techniques used in recombinant DNA technology or gene manipulation. The most frequently used methods are agarose gel electrophoresis, isolation and purification of nucleic acids, nucleic acid blotting techniques, DNA sequencing, chemical synthesis of DNA, gene transfer methods, polymerase chain reaction, construction of gene library, radiolabeling of nucleic acids etc. Gel electrophoresis is a routinely used analytical technique for the separation and purification of specific DNA fragments. PCR is a cell free amplification technique. The three – step cycle is repeated to obtain copies of target DNA in large numbers.

The DNA markers are highly useful for genetic mapping of genomes. RFLPS (Restriction Fragment Length Polymorphisms), VNTRs (mini satellites or Variable Number Tandem Repeats), STRs (Microsatellites or Simple Random Repeats), SNPs (Single Nucleotide Polymorphisms) are types of DNA sequences (stretch of DNA) which can be used as markers. These markers are used in disease diagnosis and DNA fingerprinting.

Evaluation

Multiple choice questions

1. Which of the following properties is essential for the function of a tRNA molecule?



   a. Recognition of a codon
   b. Recognition of an anticodon
   c. Ability to distinguish one amino acid from another
   d. Recognition of DNA molecule

2. Which chain termination codon could be formed by a single base change from UCG, UGG and UAU?
   a. UAA b. UAG
   c. UGA d. AUG

3. Which of the following base-pair changes are transitions?
   a. AT → TA b. AT → GC
   c. Both a and b d. GC → AT

4. UV light usually causes mutations by a mechanism involving
   a. One-strand breakage in DNA
   b. Deletion of DNA segments
   c. Induction of thymine dimers and their persistence
   d. Inversion of DNA segments

5. The form of genetic information used directly in protein synthesis is
   a. DNA b. mRNA
   c. rRNA d. tRNA

6. _____ display one anticodon each
   a. eukaryotic mRNAs
   b. transfer RNAs
   c. ribosomal RNAs
   d. mRNAs

7. _____ contains exons and introns.
   a. Eukaryotic mRNAs b. rRNA
   c. tRNAs d. primers

8. The symbol lac^+ refer to

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